

Object detection with YOLO

กิตติรักษ์ ม่วงมิ่งสุข (กิ)

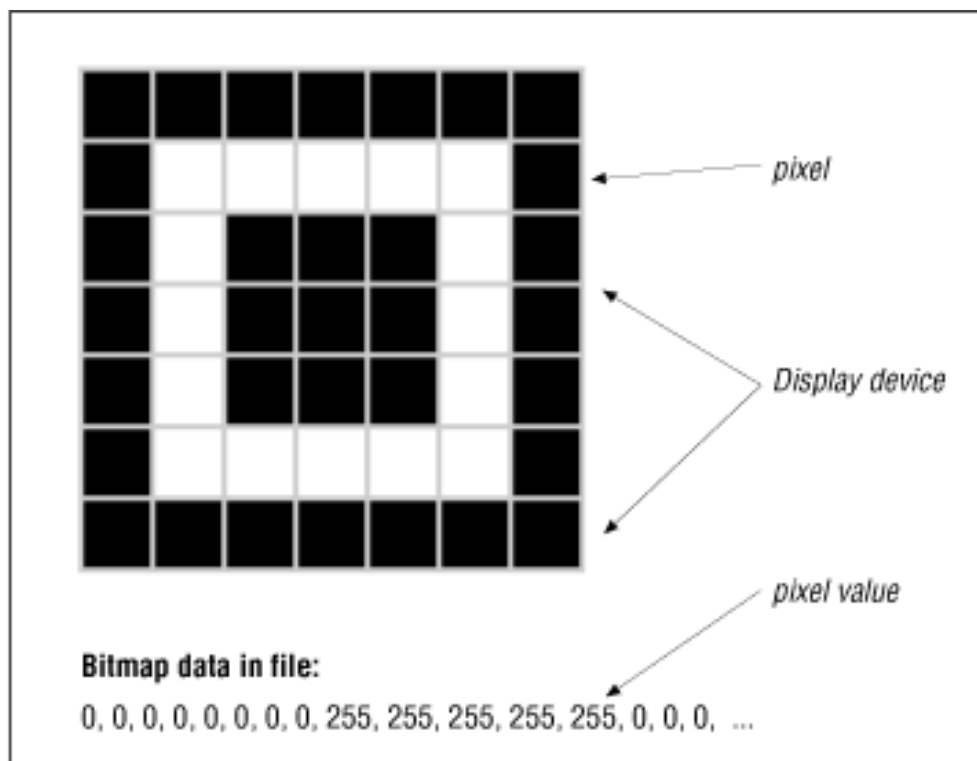
Last Modify: Feb 25, 2026

วิดีโอต้องชม

- Amazon Go
 - <https://www.youtube.com/watch?v=NrmMk1Myrxc>
- AI ชนะแพทย์ในการแข่งวินิจฉัยภาพสมอง
 - <https://voicetv.co.th/watch/r19RKvPGQ>
- “การเรียนรู้เชิงลึกอะไร | What is Deep Learning?”
 - <https://www.youtube.com/watch?v=iJq8ILbmcLU>

ความท้าทาย

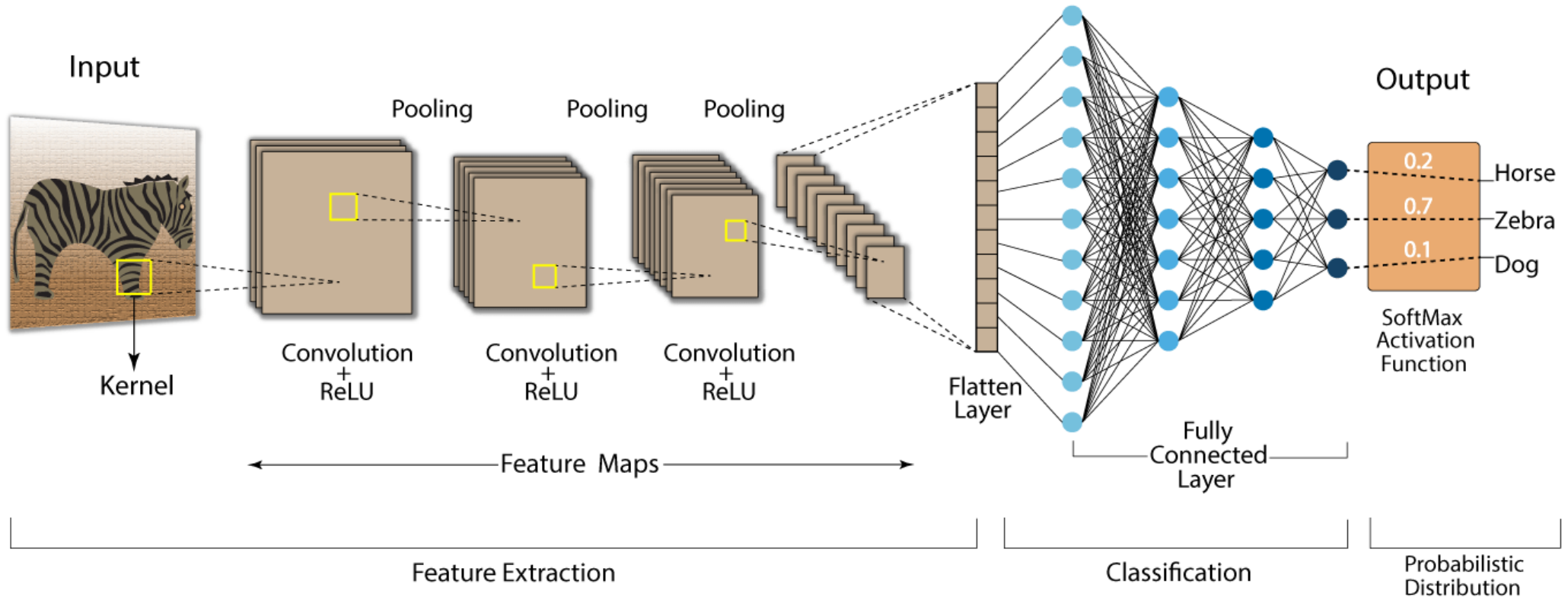
- คอมพิวเตอร์ไม่ได้เห็นภาพแบบเรา แต่เห็นเป็น “ตัวเลข”



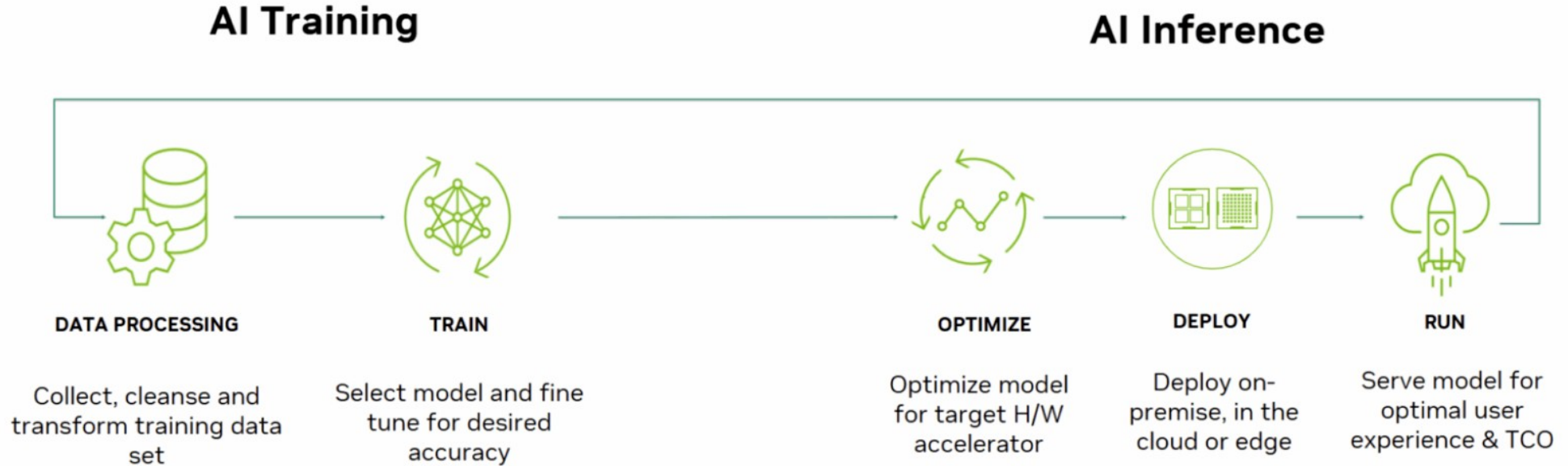
Link

- CS232 – Deep learning for computer vision (Stanford)
 - <https://cs231n.github.io/convolutional-networks/>
- Poloclub - CNN Explainer
 - <https://poloclub.github.io/cnn-explainer/>
- YOLO Architecture
 - <https://www.geeksforgeeks.org/machine-learning/yolo-you-only-look-once-real-time-object-detection/>
- Tensorflow playground <https://playground.tensorflow.org/>

Convolution Neural Network (CNN)



Methodology



What is COCO?

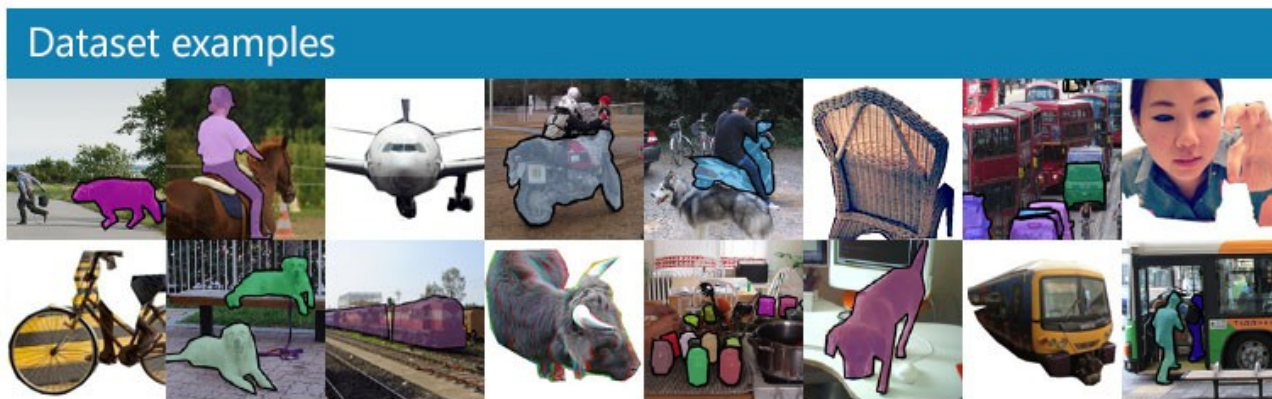


COCO is a large-scale object detection, segmentation, and captioning dataset. COCO has several features:

- ✓ Object segmentation
- ✓ Recognition in context
- ✓ Superpixel stuff segmentation
- ✓ 330K images (>200K labeled)
- ✓ 1.5 million object instances
- ✓ 80 object categories
- ✓ 91 stuff categories
- ✓ 5 captions per image
- ✓ 250,000 people with keypoints

Common Objects in Context(COCO)

- It is designed to encourage research on a wide variety of object categories and is commonly used for benchmarking computer vision models.
- <https://cocodataset.org/>



YOLO Pre-trained Models

- YOLO - You Only Live Once
- YOLO เป็นโมเดลสำเร็จรูป ของ ultralytics ใช้ทำ

- object detection
- instance segmentation
- image classification
- pose estimation
- multi-object tracking

<https://docs.ultralytics.com/>

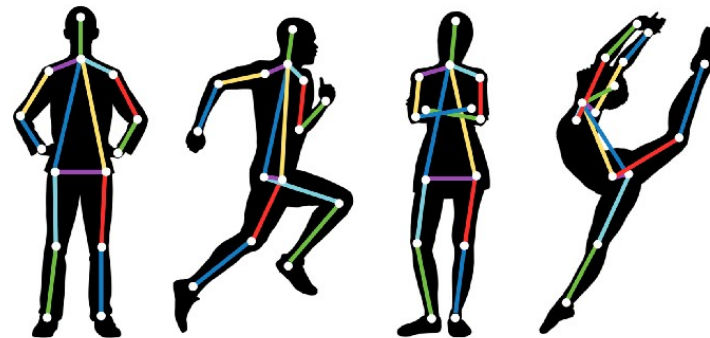


Image from: <https://kemtai.com/blog/the-complete-guide-to-human-pose-estimation/>

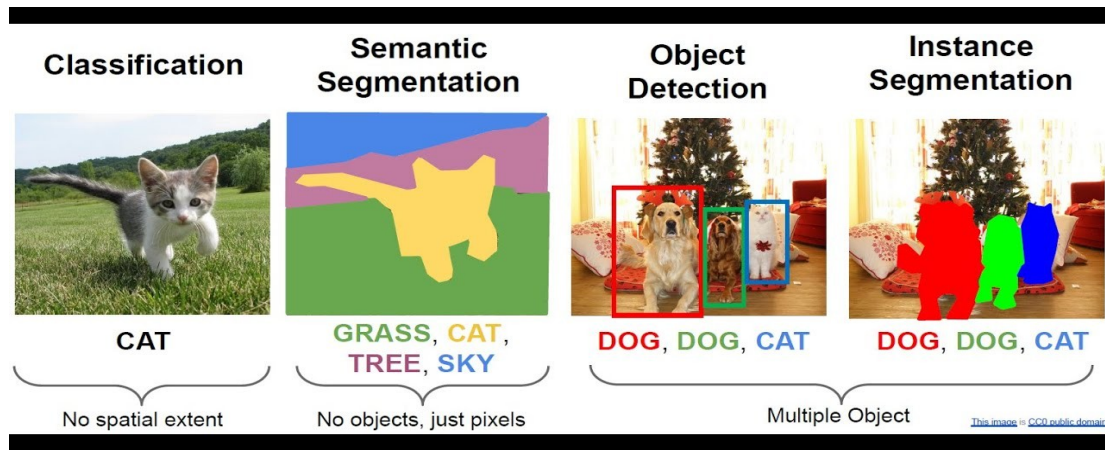


Image from: <https://keymakr.com/blog/semantic-segmentation-vs-object-detection-understanding-the-differences/>

Detect !

● Install

```
pip install ultralytics
```

● Run

```
from ultralytics import YOLO
import cv2

model = YOLO('yolov8n.pt')
results = model('image.jpg')
for r in results:
    im_array = r.plot()
    cv2.imshow('Object Detection', im_array)
    cv2.waitKey(0)
```

Workshop

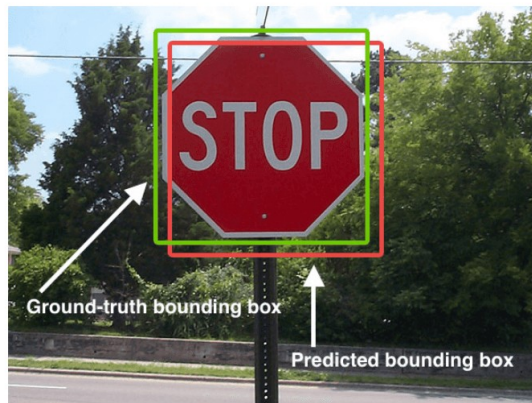
- Train model object detection ด้วยข้อมูล “ไฟ” ตามลิงค์ต่อไปนี้
 - <https://github.com/EdgeElectronics/TensorFlow-Object-Detection-API-Tutorial-Train-Multiple-Objects-Windows-10>
- Training steps
 - Data Collection: ถ่ายภาพไฟในสภาวะแสงต่างๆ
 - Annotation: ตีกรอบระบุตำแหน่ง (XML/Pascal VOC format) และแปลงเป็น YOLO format
 - Training: สอนโมเดลด้วย 50 Epochs
 - Validation: ทดสอบความแม่นยำด้วยภาพที่ไม่เคยเห็นมาก่อน

Label image tools

- Roboflow (web base, on Cloud)
 - <https://roboflow.com/>
- CVAT (Computer Vision Annotation Tool)
 - <https://github.com/cvat-ai/cvat?tab=readme-ov-file>
- LabelImg (App)
 - `pip install LabelImg`
- Label Studio (web base, on premise)
 - https://labelstud.io/guide/quick_start

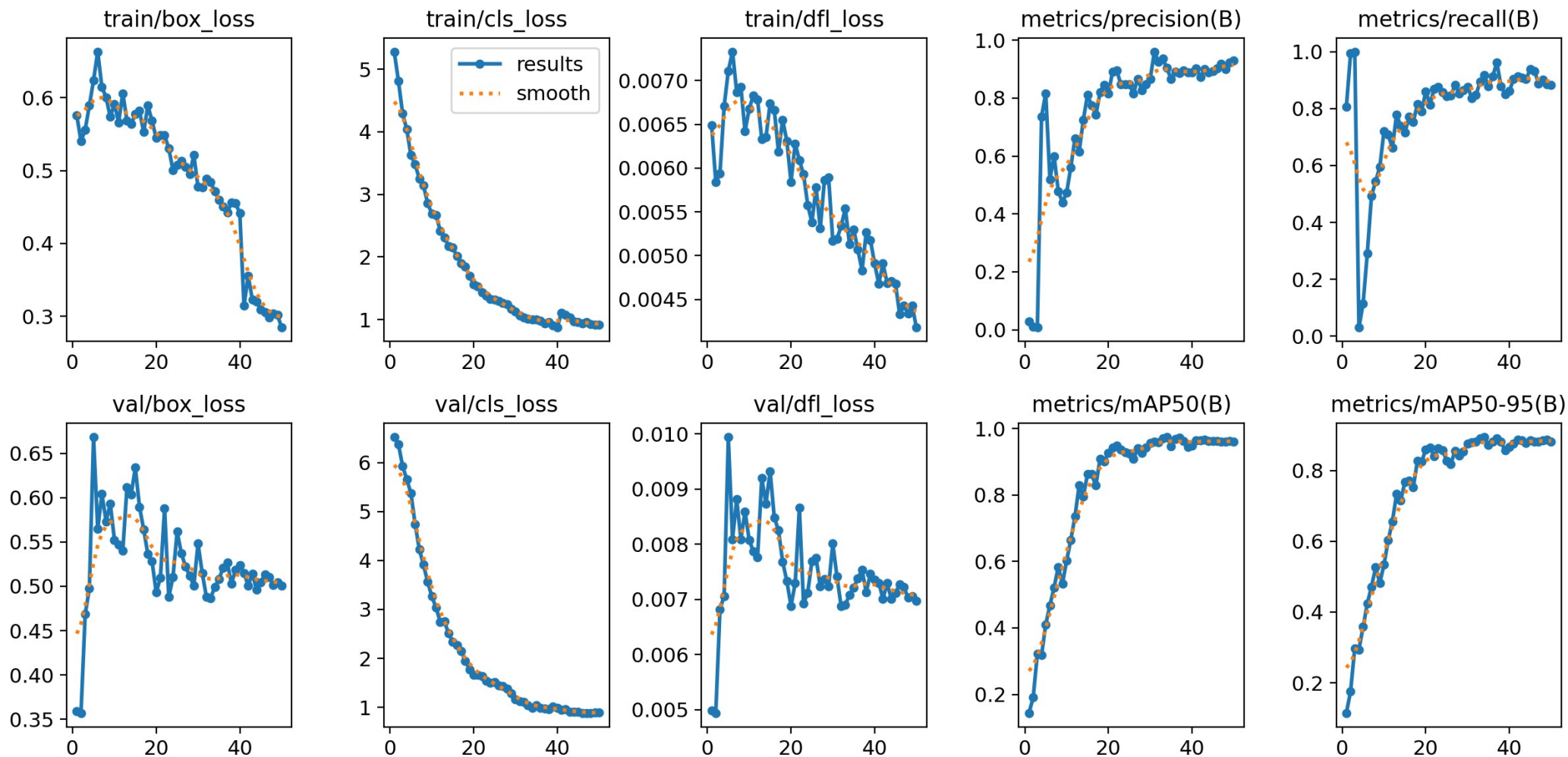
Explain essential variable

- **Epoch** คือ จำนวนครั้งที่โมเดล "อ่านข้อมูลทั้งหมด" ใน Dataset ตั้งแต่แรกจนถึงรูปสุดท้ายจนครบ 1 รอบ
- **mAP** (Mean Average Precision)
 - mAP50 (Mean Average Precision at IoU 0.5)
 - IoU 0.5 (Intersection over Union)
 - mAP50 คือการวัดความแม่นยำโดยใช้เกณฑ์ความทับซ้อนที่ 50%



$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$


Image from: <https://www.v7labs.com/blog/mean-average-precision>





Source: <https://github.com/developmentseed/label-maker/blob/master/examples/walkthrough-tensorflow-object-detection.md>

Raster vision



Object Detection



Source: <https://github.com/azavea/raster-vision-examples#how-to-run-an-example>



Q & A